Numerical Analysis

Assignment #4

Please hand in the C or Matlab code (.m files), graphics, and a brief description of your reasoning as well as comments if any.

Please upload to the 'hw4' directory if you submit your homework in time.

Problem 1:

Construct the Lagrange interpolating polynomials for the following functions, and find a bound for the absolute error on the interval $[x_0, x_n]$.

a. $f(x) = e^{2x}\cos 3x, \quad x_0 = 0, x_1 = 0.3, x_2 = 0.6, n = 2$

b. $f(x) = \sin(\ln x), \quad x_0 = 2.0, x_1 = 2.4, x_2 = 2.6, n = 2$

a. $f(x) = e^{2x}\cos(3x)$

计算函数值:

$y_0 = f(0) = e^0 \cos(0) = 1$

$y_1 = f(0.3) = e^{0.6} \cos(0.9) \approx 1.822 \times 0.6216 \approx 1.1326$

$y_2 = f(0.6) = e^{1.2} \cos(1.8) \approx 3.320 \times (-0.2272) \approx -0.7543$

构建基函数:

$L_0(x) = [(x-0.3)(x-0.6)] / [(0-0.3)(0-0.6)] = (x^2-0.9x+0.18)/0.18$

$L_1(x) = [(x-0)(x-0.6)] / [(0.3-0)(0.3-0.6)] = (x^2-0.6x)/(-0.09)$

$L_2(x) = [(x-0)(x-0.3)] / [(0.6-0)(0.6-0.3)] = (x^2-0.3x)/0.18$

拉格朗日多项式:

$P_2(x) = 1 \times L_0(x) + 1.1326 \times L_1(x) + (-0.7543) \times L_2(x)$

误差界:

首先，求 f(x)的三阶导数：$f'''(x) = e^{2x}(-46\cos(3x) - 9\sin(3x))$

然后，我们需要在区间 [0, 0.6] 上找到 $|f'''(x)|$ 和 $|W(x)| = |x(x-0.3)(x-0.6)|$ 的最大值。

通过数值计算或微积分方法，可以找到 $|W(x)|$ 在 [0, 0.6] 上的最大值约为 0.0104。

通过数值计算，$|f'''(x)|$ 在 [0, 0.6] 上的最大值发生在 x=0.6 处，约为 $|e^{1.2}(-46\cos(1.8) - 9\sin(1.8))| \approx 3.32 \times |10.45 - 8.76| \approx 5.61$.

误差界 $E \leq [\max|W(x)| \times \max|f'''(x)|] / 3! \approx (0.0104 \times 5.61) / 6 \approx 0.00973$

b. $f(x) = \sin(\ln(x))$

计算函数值:

$y_0 = f(2.0) = \sin(\ln(2.0)) \approx \sin(0.693) \approx 0.6389$

$y_1 = f(2.4) = \sin(\ln(2.4)) \approx \sin(0.875) \approx 0.7675$

$y_2 = f(2.6) = \sin(\ln(2.6)) \approx \sin(0.956) \approx 0.8169$

构建基函数:

$L_0(x) = [(x-2.4)(x-2.6)] / [(2.0-2.4)(2.0-2.6)] = (x^2-5x+6.24)/0.24$

$L_1(x) = [(x-2.0)(x-2.6)] / [(2.4-2.0)(2.4-2.6)] = (x^2-4.6x+5.2)/(-0.08)$

$L_2(x) = [(x-2.0)(x-2.4)] / [(2.6-2.0)(2.6-2.4)] = (x^2-4.4x+4.8)/0.12$

拉格朗日多项式:

$P_2(x) = 0.6389 \times L_0(x) + 0.7675 \times L_1(x) + 0.8169 \times L_2(x)$

误差界:

$f'''(x) = [\cos(\ln(x)) + 3\sin(\ln(x))] / x^3$

在区间 [2.0, 2.6] 上，$|W(x)| = |(x-2.0)(x-2.4)(x-2.6)|$ 的最大值约为 0.033。

$|f'''(x)|$ 在 [2.0, 2.6] 上的最大值发生在 x=2.0 处，约为 $|\cos(\ln(2))+3\sin(\ln(2))|/2^3 \approx (0.769 + 3\times0.639)/8 \approx 0.336$.

误差界 $E \leqslant [\max|W(x)| \times \max|f'''(x)|] / 3! \approx (0.033 \times 0.336) / 6 \approx 0.00185$

Problem 2:

Let $P_3(x)$ be the interpolating polynomial for the data $(0, 0)$, $(0.5, y)$, $(1, 3)$, and $(2, 2)$. The coefficient of x^3 in $P_3(x)$ is 6. Find y.

已知：

$x_0 = 0$, $f(x_0) = 0$

$x_1 = 0.5$, $f(x_1) = y$

$x_2 = 1$, $f(x_2) = 3$

$x_3 = 2$, $f(x_3) = 2$

一阶均差:

$f[x_0, x_1] = (y - 0) / (0.5 - 0) = 2y$

$f[x_1, x_2] = (3 - y) / (1 - 0.5) = 6 - 2y$

$f[x_2, x_3] = (2 - 3) / (2 - 1) = -1$

二阶均差:

$f[x_0, x_1, x_2] = (f[x_1, x_2] - f[x_0, x_1]) / (x_2 - x_0) = (6 - 2y - 2y) / (1 - 0) = 6 - 4y$

$f[x_1, x_2, x_3] = (f[x_2, x_3] - f[x_1, x_2]) / (x_3 - x_1) = (-1 - (6 - 2y)) / (2 - 0.5) = (-7 + 2y) / 1.5$

三阶均差:

$f[x_0, x_1, x_2, x_3] = (f[x_1, x_2, x_3] - f[x_0, x_1, x_2]) / (x_3 - x_0)$

$= [((-7 + 2y) / 1.5) - (6 - 4y)] / (2 - 0)$

又已知三阶均差（即 x^3 的系数）是 6

$6 = [((-7 + 2y) / 1.5) - (6 - 4y)] / 2$

$12 = (-7 + 2y) / 1.5 - (6 - 4y)$

$12 = (-7 + 2y - 1.5 \times (6 - 4y)) / 1.5$

$18 = -7 + 2y - 9 + 6y$

$18 = -16 + 8y$

$34 = 8y$

$y = 34 / 8 = 4.25$

答案： y 的值是 4.25

Problem 3:

Neville's method is used to approximate $f(0.4)$, giving the following table.

$x_0 = 0$	$P_0 = 1$			
$x_1 = 0.25$	$P_1 = 2$	$P_{01} = 2.6$		
$x_2 = 0.5$	P_2	$P_{1,2}$	$P_{0,1,2}$	
$x_3 = 0.75$	$P_3 = 8$	$P_{2,3} = 2.4$	$P_{1,2,3} = 2.96$	$P_{0,1,2,3} = 3.016$

Determine $P_2 = f(0.5)$.

Suppose $x_j = j$, for $j = 0, 1, 2, 3$ and it is known that

$$P_{0,1}(x) = 2x + 1, \quad P_{0,2}(x) = x + 1, \quad \text{and} \quad P_{1,2,3}(2.5) = 3.$$

Find $P_{0,1,2,3}(2.5)$.

内维尔方法的迭代公式：

$P_{i,k}(x) = [(x - x_i)P_{i+1,k}(x) - (x - x_k)P_{i,k-1}(x)] / (x_k - x_i)$

第一部分：

$x_0=0$, $x_1=0.25$, $x_2=0.5$, $x_3=0.75$。

使用 $P_{2,3}(0.4) = 2.4$

$P_{2,3}(0.4) = [(0.4 - x_2)P_3 - (0.4 - x_3)P_2] / (x_3 - x_2)$

已知 $P_3=f(x_3)=f(0.75)=8$，$P_{2,3}(0.4)=2.4$。

$2.4 = [(0.4 - 0.5) \times 8 - (0.4 - 0.75) \times P_2] / (0.75 - 0.5)$

$2.4 = [(-0.1) \times 8 - (-0.35) \times P_2] / 0.25$

$2.4 \times 0.25 = -0.8 + 0.35 \times P_2$

$0.6 = -0.8 + 0.35 \times P_2$

$1.4 = 0.35 \times P_2$

$P_2 = 1.4 / 0.35 = 4 = f(x_2) = f(0.5)$

第二部分：

确定函数值:

$x_0=0$, $x_1=1$, $x_2=2$, $x_3=3$.

从 $P_{0,1}(x) = 2x + 1$ 中得到：

$f(x_0) = f(0) = P_{0,1}(0) = 1$

$f(x_1) = f(1) = P_{0,1}(1) = 3$

从 $P_{0,2}(x) = x + 1$ 中得到：

$f(x_2) = f(2) = P_{0,2}(2) = 3$ (与 $f(x_0)=1$ 一致)

构建 $P_{0,1,2}(x)$:

先求 $P_{0,1,2}(x)$

$P_{0,1,2}(x) = [(x-x_0)P_{1,2}(x) - (x-x_2)P_{0,1}(x)] / (x_2-x_0)$

$P_{1,2}(x)$ 是过 $(x_1, f(x_1))=(1,3)$ 和 $(x_2, f(x_2))=(2,3)$ 的直线，即 $P_{1,2}(x) = 3$。

$P_{0,1,2}(x) = [(x-0)(3) - (x-2)(2x+1)] / (2-0)$

$P_{0,1,2}(x) = [3x - (2x^2 - 3x - 2)] / 2 = (-2x^2 + 6x + 2) / 2 = -x^2 + 3x + 1.$

计算 $P_{0,1,2}(2.5)$:

$P_{0,1,2}(2.5) = -(2.5)^2 + 3(2.5) + 1 = -6.25 + 7.5 + 1 = 2.25.$

计算 $P_{0,1,2,3}(2.5)$:

$P_{0,1,2,3}(x) = [(x-x_0)P_{1,2,3}(x) - (x-x_3)P_{0,1,2}(x)] / (x_3-x_0)$

代入 x = 2.5:

$P_{0,1,2,3}(2.5) = [\ (2.5-0)P_{1,2,3}(2.5) - (2.5-3)P_{0,1,2}(2.5)\ ] / (3-0)$

$P_{0,1,2,3}(2.5) = [\ 2.5 \times (3) - (-0.5) \times (2.25)\ ] / 3$

$P_{0,1,2,3}(2.5) = [\ 7.5 + 1.125\ ] / 3 = 8.625 / 3 = 2.875.$

故 $P_{0,1,2,3}(2.5) = 2.875$

Problem 4:

For a function f, the forward-divided differences are given by

$x_0 = 0.0$	$f[x_0]$		
		$f[x_0, x_1]$	
$x_1 = 0.4$	$f[x_1]$		$f[x_0, x_1, x_2] = \frac{50}{7}$
		$f[x_1, x_2] = 10$	
$x_2 = 0.7$	$f[x_2] = 6$		

Determine the missing entries in the table.

1.求 $f[x_0, x_1]$:

$f[x_0, x_1, x_2] = (f[x_1, x_2] - f[x_0, x_1]) / (x_2 - x_0)$

$50/7 = (10 - f[x_0, x_1]) / (0.7 - 0.0)$

$(50/7) \times 0.7 = 10 - f[x_0, x_1]$

$5 = 10 - f[x_0, x_1]$

$f[x_0, x_1] = 5$

2.求 $f[x_1]$:

$f[x_1, x_2] = (f[x_2] - f[x_1]) / (x_2 - x_1)$

$10 = (6 - f[x_1]) / (0.7 - 0.4)$

$10 = (6 - f[x_1]) / 0.3$

$3 = 6 - f[x_1]$

$f[x_1] = 3$

3.求 $f[x_0]$:

$f[x_0, x_1] = (f[x_1] - f[x_0]) / (x_1 - x_0)$

$5 = (3 - f[x_0]) / (0.4 - 0.0)$

$5 = (3 - f[x_0]) / 0.4$

$2 = 3 - f[x_0]$

$f[x_0] = 1$

综上缺失的项是：

$f[x_0] = 1$

$f[x_1] = 3$

$f[x_0, x_1] = 5$

Problem 5:

Determine the natural cubic spline S that interpolates the data $f(0) = 0$, $f(1) = 1$, and $f(2) = 2$. Determine the clamped cubic spline s that interpolates the data $f(0) = 0$, $f(1) = 1$, $f(2) = 2$ and satisfies $s'(0) = s'(2) = 1$.

见 problem5.c，答案为 **S(x) = x**